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LogDigitizer

USER GUIDE

Digitization of geophysical well logs

From scanned log image to calibrated digital curve

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Version 1.0.0

2026

1. Getting started

LogDigitizer converts scanned geophysical well logs into digital curves that can be exported in LAS 2.0 format. You prepare the image, calibrate its depth and curve scales, then trace the curves using semi-automatic tools and manual corrections.

This guide follows that workflow from a scanned image to an exported LAS file. Start with one image and one curve; the later chapters explain how to handle additional scales, wrap-around curves and multiple images.

1.1. Install and launch

Run the LogDigitizer installer for Windows and follow the installation instructions. Once installed, open the application from the Start menu or the desktop shortcut. You do not need to install Python separately.

Before starting, choose a scanned log with clearly readable depth references and curve-scale values. Calibration requires at least two known depth references and two known horizontal values for each curve scale.

1.2. Create a project and add images

A project keeps the imported images, calibrations and digitized curves together.

1. Select **File → New Project**, enter a project name and choose where to save it.
2. Confirm the new project. The image-selection dialog opens automatically.
3. Select the scanned image or images you want to digitize.

The imported images appear in the **Project Explorer** on the left. Select an image to display it in the viewer. You can add further images at any time using **Add Image**.

A project may contain consecutive images covering different depth intervals, or images covering the same interval but containing different curves. Each image is prepared and calibrated individually; curves can then be combined during export.

Use **Save Project** or **Ctrl+S** regularly. To continue an earlier session, open its `.logproj` file or select it from **Recent Projects**.

Moving a project: Keep the `.logproj` file and its associated project folders together when copying the project to another location or computer.

1.3. Find your way around

The workspace has three main areas:

- **Project Explorer — left:** select an image, a curve and the task you want to perform.

- **Image viewer — centre:** inspect the scanned log, place calibration references and digitize curves.
- **Task Panel — right:** access the controls for **Preprocessing**, **Calibration** or **Digitization**.

Use the mouse wheel to zoom around the pointer. To move along the image, activate **Pan** and drag with the left mouse button. Press **F** to fit the image to the viewer, or **1** to display it at its original pixel scale.

The floating readout in the viewer shows depth and curve value as you move the pointer over a calibrated image. These values describe the pointer position. Over a saved digitized section, the readout uses that section's scale and offset, and warns if they differ from the active settings in the Task Panel.

Hover over an unfamiliar icon to see its function. The information buttons beside the task controls provide additional guidance.

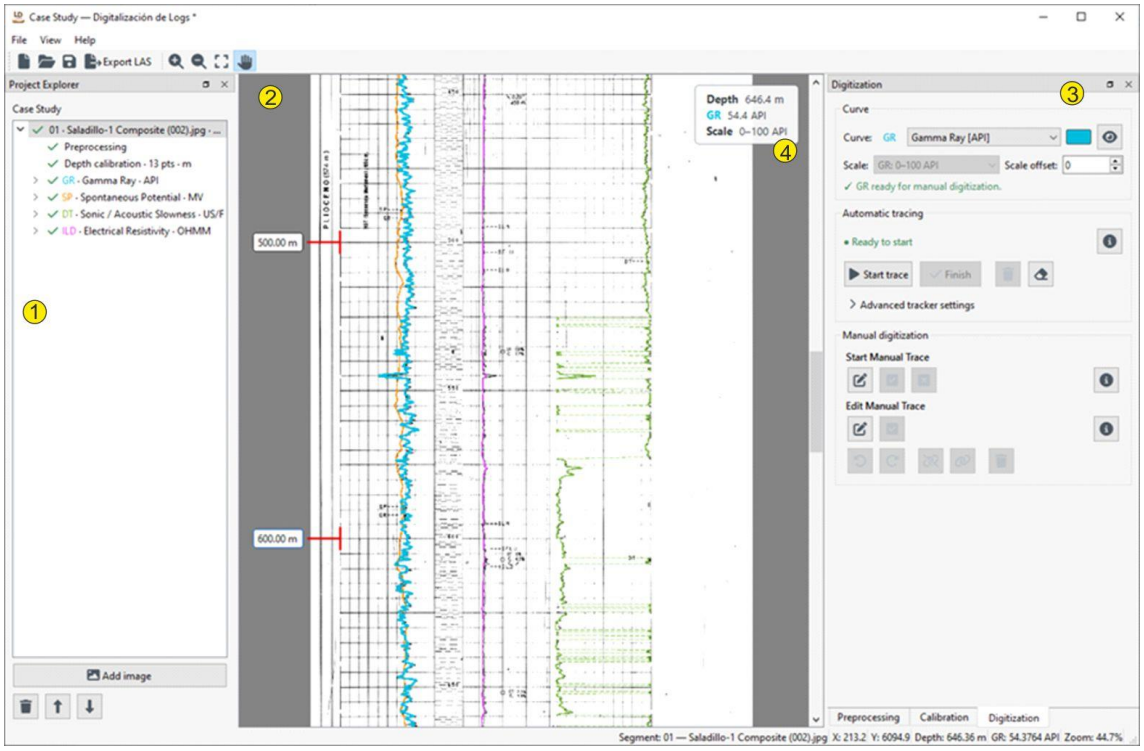


Figure 1.1. The LogDigitizer workspace: (1) Project Explorer, (2) image viewer, (3) Task Panel and (4) floating depth and curve-value readout.

1.4. Follow the workflow

For your first curve, work through these stages:

Stage	What you do
Prepare	Rectify the image and adjust its appearance where needed.
Calibrate	Relate image positions to known depths and curve values.

Digitize	Trace the curve semi-automatically, review the result and correct difficult intervals manually.
Export	Select the curves and depth sampling interval, check the coverage and save the LAS file.

Complete image preparation before calibration: preprocessing becomes locked once an image contains calibration or digitization data.

The next chapter explains how to prepare your first image and calibrate its depth and curve scale.

2. Prepare and calibrate the image

Before tracing a curve, prepare the scanned image and relate its positions to known depth and curve values. Accurate calibration is essential: a trace may follow the printed curve perfectly and still produce incorrect data if its references or units are wrong.

Select the image in the **Project Explorer**, then open **Preprocessing**.

2.1. Rectify the image

Rectification corrects the geometry of a tilted or distorted scan using four corners around the log area. This step is required before calibration.

1. Activate **Define 4 corners** and place the corners on clear references defining the rectangular log area.
2. Zoom in and adjust their positions where necessary.
3. Open **Preview** and inspect the alignment of the log boundaries and grid.
4. When the result is satisfactory, select **Apply Rectification**.

The corners define the reference geometry; they do not crop the scan. The complete image is retained in the corrected result.

If the lateral boundaries remain curved, use **Multipoint correction** before applying the rectification. Select **Add pair**, then place a point on each boundary at the same transverse reference on the log. Repeat only where additional alignment is needed, and inspect the preview again.

Review the geometry before selecting **Apply Rectification**: the corner and multipoint controls become unavailable after it is applied.

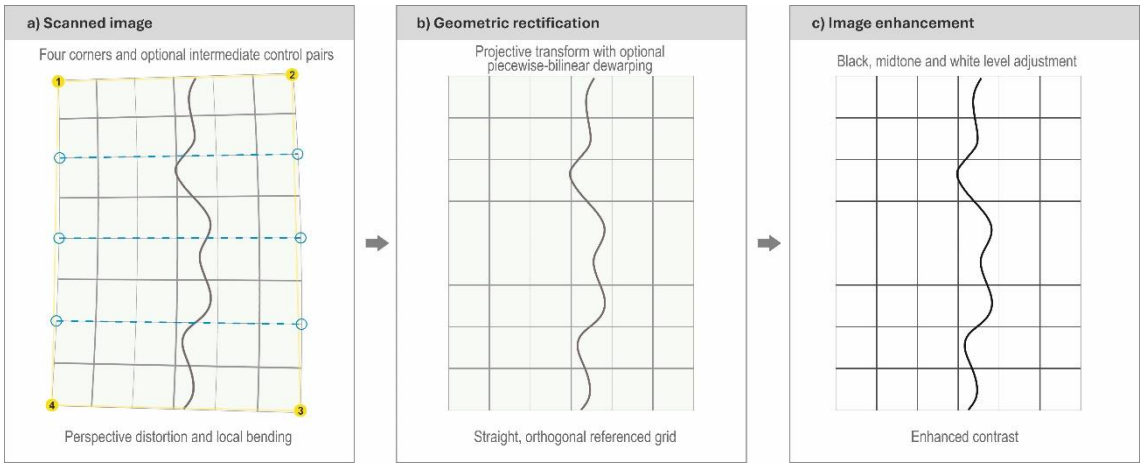


Figure 2.1. Image preprocessing in LogDigitizer. (a) Scanned log with four corner references (yellow) and optional intermediate control pairs (blue). (b) Rectified image after correcting tilt and local distortion. (c) Enhanced image after adjusting black, midtone and white levels to improve curve visibility.

2.2. Improve image clarity

After rectification, use **Image Enhancement** if the curves are faint or the paper background is uneven. Adjust **Black**, **Midtones** and **White** while inspecting the image, then select **Apply Enhancement** to retain the changes.

Aim for a clearly distinguishable curve while preserving thin lines and faint intervals. Strong contrast adjustments can remove useful detail or make the curve harder to separate from the grid.

Enhancement is optional and can be revisited at any time, including during digitization. Different parts of a scanned log may require different adjustments, so you can return to **Image Enhancement** and modify the settings to suit the interval you are working on. These adjustments affect the image as a whole, rather than only the visible area.

2.3. Calibrate depth

Open **Calibration** and use **Depth calibration (Y)** to connect vertical image positions to borehole depth.

1. Select the depth unit, **m** or **ft**, before adding the first point.
2. Activate **Add Y Point**, click a clearly identified depth reference and enter its value.
3. Repeat at a second reference, preferably well separated from the first and spanning the interval you intend to digitize.
4. Add further references where needed to account for variations in vertical spacing, then deactivate the capture tool.

For example, if two labelled grid lines represent 100 m and 150 m, place a calibration point precisely on each line and enter those depths. Use the grid line itself rather than the position of its printed label.

Depth values must increase or decrease consistently along the image. The panel indicates when the calibration is valid.

Check the result against another labelled depth reference, if available, using the floating readout in the viewer. If it does not agree, review the point positions, entered values and selected unit.

2.4. Define the curve and calibrate its scale

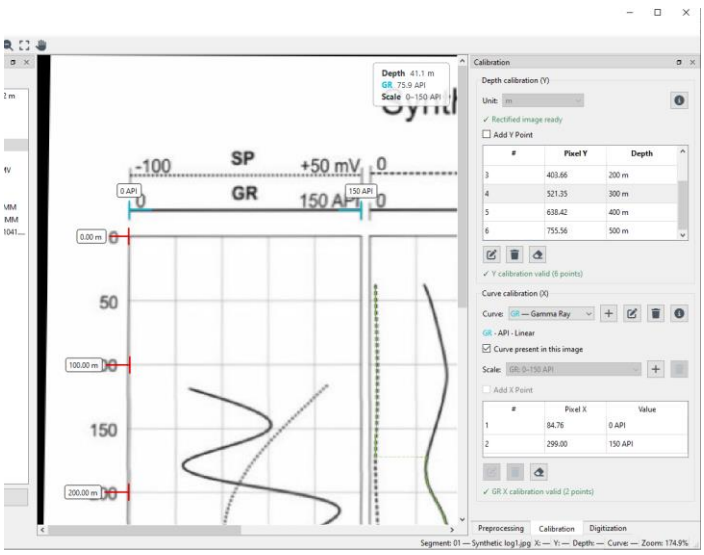
In **Curve calibration (X)**, create or select the curve you want to digitize. When creating a curve, choose a predefined parameter or enter its details. Check the mnemonic, unit and whether the printed scale is **Linear** or **Logarithmic**.

For this first example, consider a gamma-ray curve identified as **GR**, with a linear scale from **0 to 100 API**.

1. Select the curve and ensure **Curve present in this image** is enabled.
2. Activate **Add X Point**, click the reference line corresponding to **0 API** and enter `0`.
3. Click the reference line corresponding to **150 API** and enter `150`.

Use references belonging to that curve's printed scale. The two horizontal positions and their values must differ; logarithmic scales require positive values.

Each curve is defined once for the project, but its horizontal scale must be calibrated separately in every image where it appears. Reuse the existing curve definition when moving to another image.



Additional scales and wrap-around curves are covered in Chapter 4.

Figure 2.2. Depth and curve-scale calibration for a gamma-ray log. Depth markers coincide with labelled horizontal grid lines; X-calibration markers identify the 0 and 150 API references. The Calibration panel shows the selected curve and valid calibrations.

2.5. Check before tracing

Before opening **Digitization**, check that:

- Depth readings agree with the printed references and use the correct unit.
- The selected curve, unit and scale type match the log.
- Curve-value readings agree with the horizontal scale.

The viewer readout reports the value corresponding to the pointer position; you do not need a digitized trace to check calibration.

To refine a reference, drag its marker. To correct an entered value, double-click its label. Vertical position determines depth calibration; horizontal position determines curve-scale calibration.

Once these checks are complete, the image is ready for tracing.

3. Digitize and correct curves

Use semi-automatic tracing to capture the curve, then correct difficult intervals manually. Both methods contribute to the same output curve, so you can combine them as needed.

Select the image and open **Digitization**. Check the active **Curve**, **Scale** and **Scale offset** before starting. For a curve using its normal printed range, leave the offset at **0**.

3.1. Start a semi-automatic trace

In **Automatic tracing**, select **Start trace** and click precisely on the curve near the top of the interval you want to digitize. Choose a clear point away from grid intersections or annotations.

The tracker follows the curve downwards through the image and produces a temporary draft. Inspect it as you move along the log, checking that it follows the intended curve rather than a grid line or a neighbouring trace.

The draft remains available for correction until you select **Finish**.

3.2. Correct or resume the draft

If the tracker leaves the curve, click the correct curve at the depth where you want tracking to restart. The draft above that depth is retained, and the lower part is replaced by a new trace from your selected point.

Choose the correction point near the beginning of the deviation so that an incorrect interval is not left above it.

If tracking stops at an interruption, click the curve where it becomes clear again to resume. Inspect the connection across the interruption; use a manual section later if it needs a more accurate reconstruction.

Repeat these corrections as necessary, then select **Finish** to retain the trace as a saved section. Use the draft's trash icon if you want to discard it and start again.

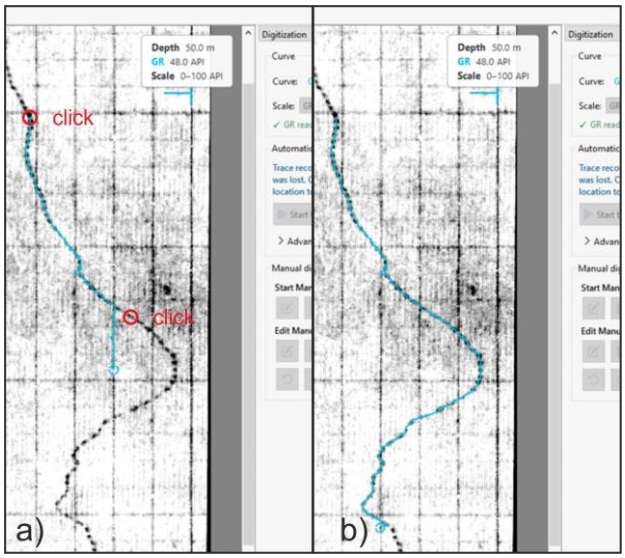


Figure 3.1. Correcting a semi-automatic trace. (a) The tracker leaves the curve and follows a grid line. A new point is selected on the correct curve near the start of the deviation. (b) Tracking resumes from that point, preserving the upper part of the draft.

3.3. Adjust tracking when needed

Start with the default settings. If the same problem occurs repeatedly, open **Advanced tracker settings** and adjust the control most relevant to it.

Problem	Adjustment to try
A faint curve is not detected reliably	Increase **Sensitivity**.
The trace jumps to nearby lines	Reduce **Search width**.
Abrupt lateral changes are missed	Increase **Search width**.
Sharp bends are not followed	Increase **Flexibility**.
Short breaks repeatedly stop tracking	Increase **Gap tolerance**.

Change one setting at a time and review the result. Higher values are not always better: a wider search or greater gap tolerance can also connect unrelated features.

Finish or discard the current draft before changing tracker settings. If poor image contrast is the main difficulty, return to **Image Enhancement**, adjust the image and apply the changes before trying again.

For a short, ambiguous interval, manual correction is often quicker than repeated adjustments.

3.4. Complete difficult intervals manually

Use **Manual digitization** where the curve crosses another line, becomes faint or requires a more carefully controlled reconstruction. You can also digitize an entire curve manually.

1. Check the curve, scale and offset, then select **Start Manual Trace**.
2. Place nodes along the printed curve, progressing consistently in one vertical direction.
3. Inspect the curve drawn between the nodes and select **Finish Trace** when it follows the intended path.

At least two nodes are required. Use fewer nodes along smooth intervals and add more around bends or rapid changes.

To correct a saved automatic trace, start the manual section just before the error and end it just after, placing both endpoints on reliable parts of the curve. **The manual section replaces the automatic result within its depth interval**, while the automatic data outside it are preserved. You do not need to erase the automatic interval first.

Use the same scale and offset as the interval being corrected. The floating readout helps identify the settings assigned to the saved section.



Figure 3.2. Manual correction of an automatic result. Left: an incorrect trace at a curve intersection. Right: control nodes define a replacement section that follows the intended curve and connects with the reliable automatic intervals above and below.

3.5. Refine saved sections

To refine a manual section, activate **Edit Trace** and select it in the viewer. Its control nodes become visible:

- Drag a node to adjust the curve.
- Click between existing nodes to insert another.
- Right-click a node to remove it.

Select **Done Editing** when finished. Automatic sections do not expose editable nodes; correct them with a manual replacement or retrace the affected interval.

To remove an unwanted part of a saved automatic trace, activate **Erase automatic range**, start dragging directly on the trace and drag vertically across the interval. Release to apply the removal. Manual corrections within that range are preserved.

Use **Undo** (**Ctrl+Z**) or **Redo** (**Ctrl+Shift+Z**) to revise digitization edits. This history applies to the current image and is cleared when you switch images; discarding an unfinished draft cannot be undone.

3.6. Review the completed curve

Inspect the full digitized interval before moving on. Pay particular attention to curve intersections, sharp peaks, faint sections and the connections between automatic and manual sections.

Check that the trace follows the intended curve, that no unwanted gaps remain and that each section uses the correct scale and offset. Where the original curve cannot be interpreted reliably, leave a gap rather than inventing a continuation.

Finish any active draft and save the project. You can then select another curve or image, or proceed to export.

The next chapter explains how to work with additional scales, wrap-around curves and multiple images.

4. Work with scales and multiple images

Some logs use more than one horizontal scale or continue across several scanned images. Before digitizing these cases, identify how the curve is represented: a change in its printed range requires an additional scale, while a curve that wraps around the track boundary requires a scale offset.

4.1. Use an additional scale

A curve may use two printed ranges, such as **GR 0–100 API** and **GR 0–1000 API**. Define these as separate scales belonging to the same curve.

In **Calibration**, select the curve and complete its first X calibration. Then use **Add Scale** and place two X-calibration points for the additional range. Each scale is named automatically from its calibrated values and unit.

In **Digitization**, select the scale corresponding to the interval you are tracing. When the printed range changes:

1. Finish the current section.
2. Select the appropriate **Scale**.
3. Start a new section using that scale.

Each saved section retains its assigned scale. Selecting another scale for subsequent tracing does not change previously saved sections.

4.2. Digitize a wrap-around curve

On some analogue logs, the recording pen reaches one edge of the track and continues from the opposite edge. The apparent jump represents a continuation beyond the printed range.

Use **Scale offset** to recover the actual values. For a linear scale from **0 to 100 API**:

Scale offset	A position reading 30 API on the printed scale represents…
0	30 API
+1	130 API
+2	230 API
−1	−70 API

Choose the offset from the log's recording convention and the continuity of the curve.

At a wrap-around transition, finish the current section, change **Scale offset** and start another section on the opposite side of the track. Return the offset to **0** when the curve returns to its base range.

Do not trace directly across the horizontal jump. Separate sections allow each part to retain the correct offset. Scale offsets are available for linear scales.

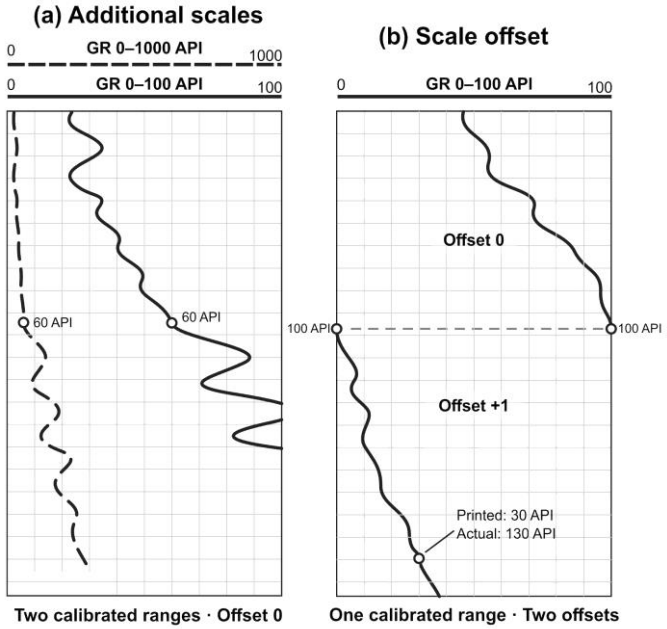


Figure 4.1. Additional scales and scale offsets. (a) A curve uses two independently calibrated printed ranges. (b) A curve wraps around a single printed range; separate sections with offsets of 0 and +1 preserve its numerical continuity.

4.3. Check a saved section's scale and offset

Move the pointer over a saved section and inspect the floating readout. It uses the scale and offset assigned to that section and warns when they differ from the active settings in the Task Panel.

Check this context before adding a manual correction. For example, a section saved with offset **+1** still requires that offset even if the panel currently shows **0**.

For an editable saved section, use **Edit Trace**, select the section, adjust **Scale** or **Scale offset**, then select **Done Editing**. Review the resulting values before continuing.

4.4. Continue a curve across consecutive images

When a long log has been scanned in parts, add the images to the same project and arrange them in their intended order in the **Project Explorer**.

Prepare and calibrate each image independently. Reuse the same project curve wherever it continues, and keep **Curve present in this image** enabled for those images.

For overlapping images:

1. Calibrate depth and the curve scale in both images.
2. Select the curve and locate the red dashed overlap guide.
3. Extend digitization slightly above or below the guide, following its displayed instruction.
4. Finish the trace before switching to the neighbouring image, then follow its corresponding guide.

The guides identify where coverage should extend so that the curve can continue between images. They are references for digitization, not cut lines or editable points.

Overlap guides are calculated **for the selected curve**. An image participates only when that curve is marked as present, so different curves may have different neighbouring images.

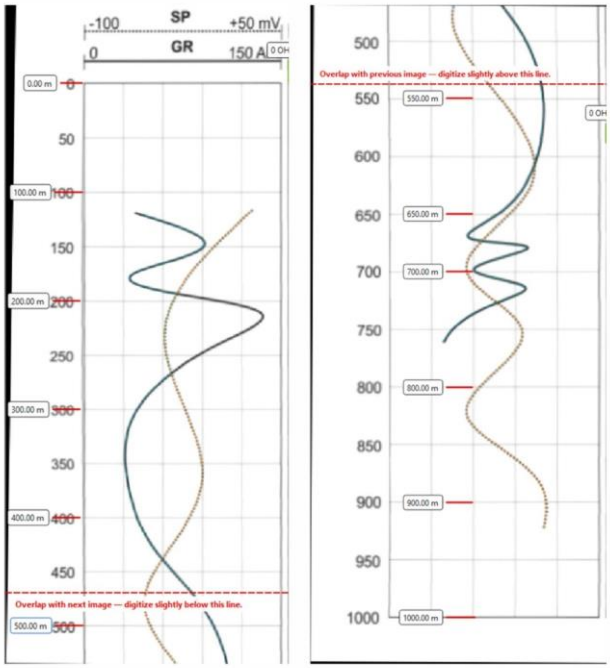


Figure 4.2. Continuing a curve across overlapping images. The red dashed guides indicate where digitization should extend in each image. Following both instructions provides coverage across the transition.

4.5. Combine images containing different curves

Images do not have to form a single vertical sequence. For example, one set may contain **GR** and **SP**, while another contains **DT** over the same depth interval. These can all belong to one project and be exported together.

Create each curve once, then indicate where it appears using **Curve present in this image**. Calibrate depth in every image and calibrate the horizontal scales only for the curves present there.

For example:

Image	Curves marked as present
Image A	GR and SP
Image B	DT

Image B does not affect the overlap guides for GR or SP. The same curve-specific approach is used when checking gaps for LAS export.

Before exporting, review the coverage of each curve separately. A complete depth interval for one curve does not mean that the others contain data throughout that interval.

5. Export and check the results

LogDigitizer exports the selected curves to a **LAS 2.0** file for use in well-log interpretation and other geoscientific applications. Sections belonging to the same curve are combined across images according to their calibrated depths.

Finish any active trace and save the project before opening **File → Export LAS…**.

5.1. Select the curves

In **Data and sampling**, select the curves you want to include. Check their mnemonics, units and digitized ranges.

Only curves with valid calibration and saved digitized data can be exported. If a curve is unavailable, hover over its row to see the reason, then return to calibration or digitization as needed.

All selected curves must use the same depth unit. Images calibrated in metres and feet cannot contribute to the same LAS export.

5.2. Choose the sampling interval

STEP defines the regular depth interval between successive samples in the exported file. For example, a STEP of **0.10 m** produces one sample every ten centimetres.

Choose an interval appropriate to the detail visible in the original log and the requirements of your subsequent analysis. A smaller STEP produces more samples, but **does not increase the accuracy or resolution of the source image**. An excessively large STEP may miss narrow features.

The export dialog displays the approximate **Pixel resolution**, expressed as depth per image pixel. Use this as a reference when choosing STEP. You can export at a finer interval, but the additional samples represent interpolation rather than new information.

LAS START and **LAS STOP** are calculated automatically from the selected curves and STEP. Review these values and the estimated number of rows before continuing.

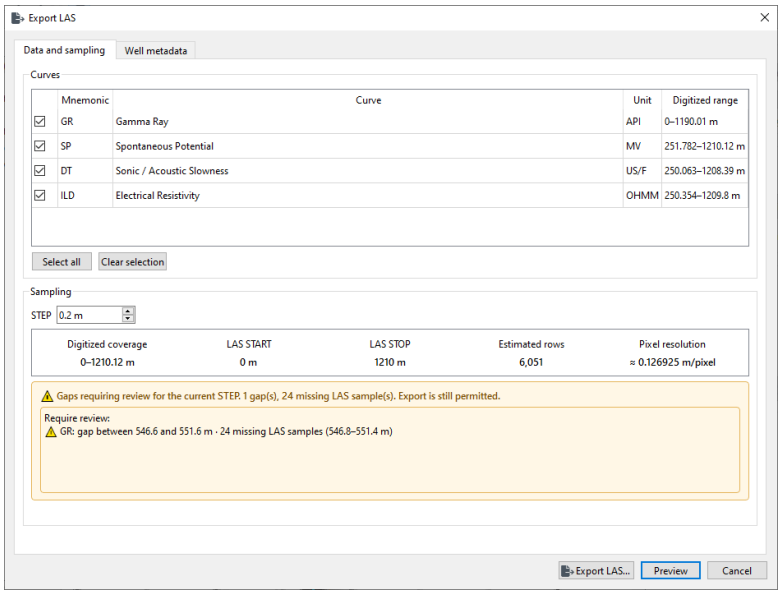


Figure 5.1. LAS export settings. Select the curves, choose the depth sampling interval (STEP), and review the output depth range, estimated row count and approximate pixel resolution.

5.3. Review coverage and gaps

Check the coverage of each selected curve, particularly where tracing was interrupted or continued in another image. Curves included in the same LAS file do not need to cover identical depth intervals.

Missing samples are written using the LAS null value, **−999.25**. This indicates absent data, not a measured curve value.

Review any **Gap QC** warnings before exporting. Coverage is assessed separately for each curve: an overlapping image containing GR cannot fill a missing interval in DT.

If a gap results from unfinished digitization, return to the image and complete the trace. If the original record is missing or unreadable, retain the gap rather than adding an unsupported continuation.

5.4. Add well information and export

Open **Well metadata** and review the borehole name and other available identification details. Enter information from the original log where possible, and leave unknown fields blank. Use **Other information** for relevant notes about the source or limitations of the digitization.

Select **Preview** to inspect the LAS text generated from the current export settings, including the well information, curve definitions and sampled data. The preview is read-only: it does not save a file or modify the project metadata. Close it to make any necessary adjustments.

Review the export summary, resolve any blocking errors, then select **Export LAS…** and choose the filename and location.

The STEP and metadata used for a successful export are retained in the project. Save the project again to preserve these settings for your next session.

5.5. Check the exported file

Open the LAS file in your usual well-log application and plot the curves against depth. Check that:

- The expected curves and units are present.
- The depth range and sampling interval are correct.
- Distinctive peaks and changes occur at the expected depths.
- Scale changes, wrap-around transitions and joins between images have plausible continuity.
- Missing intervals are recognised as null data.

Compare a few distinctive features with the scanned log, especially near corrected intervals and image boundaries.

If you find an error, return to the LogDigitizer project, correct the calibration or digitized section and export again. Keep the project alongside the LAS file so that the result can be reviewed and revised later.